

JUNIOR CERTIFICATE

PHYSICS

FORM 2

Study Notes

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UNIT 1: THERMAL EXPANSION

- **Thermal expansion** refers to a tendency for substances to increase in size when heat is added to them.
- When matter is heated, it expands and when cooled it contracts.
- When a substance is heated, its temperature rises.
- The rise in temperature increases kinetic energy of its particles. The particles move further apart and volume increases hence expansion.
- **Expansion** is the increase in size on heating of a substance.
- On the other hand, when a substance is cooled its temperature decreases.
- **Contraction** is the decrease in size on cooling of a substance.

1. Thermal Expansion in Solids

- ❖ When a solid is heated, its particles gain kinetic energy and so they vibrate with large amplitude about the fixed position.
- ❖ The vibrations weaken the IMF causing atoms to push each other away with greater forces; hence the substance expands as its volume increases.
- ❖ It may be difficult to see expansion of solids with our naked eyes but in reality expansion does occur.

a. Rate of Expansion of Solids

- ✓ Different solids expand with different amounts when heated equally.
- ✓ For example, when a bottle top of a soft drink is difficult to be removed, heat the bottle top intensively and eventually the bottle top is removed without the use of a bottle opener.
- ✓ This observation also shows that metals expand faster than non-metals, in this case, glass.
- ✓ Different solids expand with different amounts when heated equally.

2. Thermal Expansion in Liquids

- ✓ When liquid is heated its particles gain heat energy hence they vibrate vigorously.
- ✓ The vibrations weaken the IMF causing the liquid particles move far from each other, hence expansion.
- ✓ The IMF is weaker in liquids than in solids hence particles of liquid move freely unlike solid particles that only move about fixed positions.

Rate of Expansion of Liquids

- ✓ Thermometers contain a liquid (**mercury or alcohol**).
- ✓ When the thermometer is placed in cold water, the liquid contracts and the reading on the thermometer is low.
- ✓ However, when the thermometer is transferred to hot water, the liquid expands and the reading is high.
- ✓ This shows that the liquid expands when heated and contracts when cooled.
- ✓ Different liquids expand with different amounts when heated by the same amount of heat energy due to their differences in the size of the molecule.

3. Thermal Expansion in Gases

- ✓ When heated, the molecules of gas gain kinetic energy hence they vibrate.
- ✓ The vibrations weaken the IMF.
- ✓ Since gases have weak IMF, the vibrations cause the gas molecules to move far apart causing volume to increase hence expansion.
- ✓ Gases expand more than liquids (and liquids expand more than solids) and solids when heated equally because their molecules move far apart on heating.
- ✓ **Besides, Liquids and solids expand by different amounts because they have different strengths of IMF (intermolecular forces).** Those with weaker IMF expand faster than those with stronger IMF.

Real life effects and applications of thermal expansion

1. Separating stuck tumblers or glasses –

- ✓ two glasses that are stuck can be separated by wrapping the outer glass in a warm cloth.
- ✓ The heat in the cloth is transferred to the outer glass by conduction making it expands on receiving the heat hence separation.

2. Removing tightly fixed bottle covers –

- ✓ To remove or open tightly fixed bottle tops immerse the cover in hot water.
- ✓ The metal lid expands and loosens before the glass bottle does.

3. Loose fittings of electrical cables

- ✓ Telephones and electricity cables are loosely fitted between the poles to give room for expansion and contraction as the weather changes.

4. Gaps in railway tracks

- ✓ Railways tracks are laid with small but adequate gaps between them to allow room for expansion and contraction of the rails when the temperature changes.

5. Expansion gap in fences

- ✓ Expansion joints or gaps are usually created in walls to allow room for expansion of the bricks or blocks.
- ✓ They break the walls into segments to prevent the cracking caused by changes in temperature and moisture expansion.

6. Shrink fitting

- ✓ used to produce a tight fit between two pieces of metals.
- ✓ An axle just too large for hole in a wheel is cooled in liquid nitrogen at around -196°C . The cold axle fits into the wheel.
- ✓ On warming the axle expands to give a very tight fit.

7. Metal riveting

- ✓ involves joining two or more metals.
- ✓ Holes are made on the metal plates to be joined.
- ✓ A piece of a metal is then heated to red hot and fitted onto the hole.
- ✓ The piece of metal (a rivet) is hammered flat onto the metal.
- ✓ After cooling the joined metals contract and in the process pulls the two plates into a tight fix.

8. Bimetallic Thermostat

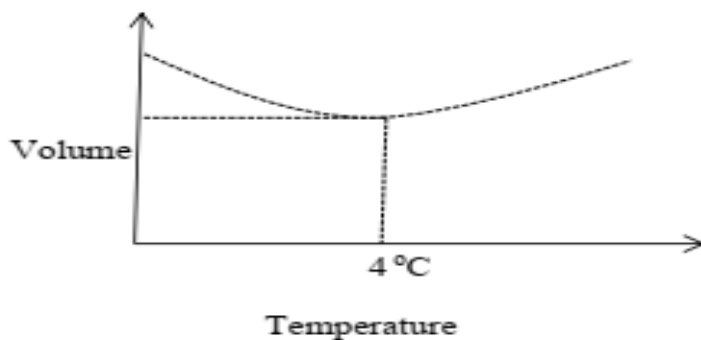
- ✓ Used to make an automatic switch.
- ✓ The thermostat is a switch using the expansion and contraction of the bimetallic strip.
- ✓ It is used to regulate or control the temperature in electric iron, electric water heater, refrigerators and fire alarms. When the temperature rises, the strip bends.
- ✓ This causes the contacts to separate and cuts off the current in the circuit. When the temperature falls, the strip contracts and straightens.
- ✓ This makes the contact, and the current is switched on again.

The effects of sudden expansion and contraction of an object

- ✓ Freezing of water in pipes often causes pipes to burst
- ✓ The expansion of the air inside a car or bicycle tyres can cause bursting of the tyres on a hot day.
- ✓ The sudden uneven expansion produced when hot water is poured into cold glass can cause the glass to crack.

Unusual / anomalous / abnormal expansion of water

- ✓ Liquids generally expand when their volume increases.
- ✓ Unlike other liquids, water expands when it cools and freezes into solid ice as its temperature drops from **100°C to 4°C**.
- ✓ Water behaves in an unusual or abnormal manner in that it contracts (**volume decreases**)
- ✓ when temperature rises from **0°C to 4°C**. **Above 4°C**, water expands with rising temperature like other liquids as shown in the graph below.



- ✓ The unusual (**abnormal**) behavior of water when heated from **0°C to 4°C** is referred to as **anomalous or abnormal expansion of water**.
- ✓ It is called **anomalous expansion of water** which is due to the **particular shape of water molecules and the direction of the forces between them**.

UNIT 2: DENSITY

- ✓ **Density** refers to mass of a given substance per unit volume.
- ✓ The formula for density is; $\rho = m/v$, where ρ (**rho**) = density in Kg / m³ or g/ cm³, m = mass in kg or g and v = volume in m³ or cm³.
- ✓ The **SI Unit** of density is kilogram per cubic metre (**Kg/m³**). The other unit of density is grams per cubic centimeter (**g/cm³**).

1) Determining the density of a regular solid

A regular solid is the one that has uniform cross – section (e.g. cuboids).

- Find the mass of the regular solid using a triple beam balance and record it.
- Measure length, breadth and height of the solid using a ruler and record them.
- Find volume by multiplying length, breadth and height
- Find density by dividing mass by volume i.e. density = mass / volume

2) Determining the density of an irregular solid

An irregular solid is the one that has no uniform cross sectional area and flat surfaces (e.g. a small stone)

Apparatus

- An irregular stone
- A beam balance
- Measuring cylinder with water

Procedure

- Measure the mass (m) of the stone using a beam balance
- Determine the volume (V) of the stone using displacement method

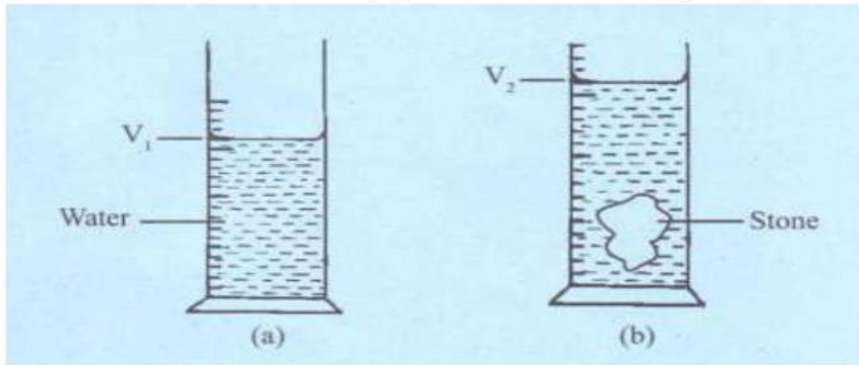


Fig. 3.1: Measuring volume of a stone

$V = V_1 - V_2$ where V_1 is the initial volume

V_2 is the final volume

$$V = V_1 - V_2$$

- Determine the density (ρ) of the iron stone.

$$\text{Density} = \frac{\text{mass (m)}}{\text{volume (v)}}$$

$$= \frac{m}{v}$$

Conclusion

Using this method, the density of a solid is given by:

$$\begin{aligned} \text{Density} &= \frac{\text{mass}}{\text{volume}} \\ &= \frac{m}{v} \end{aligned}$$

Relating density Of substances to their states

- ✓ Density of a Substance (Liquids, Solids and Gases) depends on the arrangement of its particles.
- ✓ Generally solids are denser than both liquids and gases because the particles in solids are closely packed.

- ✓ There are many particles per unit volume in a solid state as compared to the liquid and gaseous states.
- ✓ In liquids there are spaces between the particles since the particles are slightly further apart.
- ✓ Hence the density of most liquids is slightly lower than that of solids. Particles in a gas are far apart.
- ✓ This results into gases having the lowest densities comparatively.

Relative density of a Substance

- ✓ It is defined as the ratio of density of a substance to density of water.
- ✓ This gives how many times a substance is denser than water.
- ✓ Remember that the density of water **at 4°C is 1g/cm³**.
- ✓ Relative density as a ratio has no units.
- ✓ Since the density of water at 4°C is 1g/cm³; then we conclude that relative density is numerically equal to the density of the substance.

Relative density = Density of substance / Density of water at 4°C

Average density of a Substance

- ✓ Average density of a substance refers to the ratio of total mass of the mixture to the total volume of the mixture.
- ✓ **Mathematically,**
Average density = Total mass of the mixture / total volume of the mixture.
- ✓ For example, the density of sugar solution is an average density of sugar and water.
- ✓ The average density influences the behavior of the substance in water.
- ✓ For instance, a fresh egg sinks in water because its average density is greater than the density of water which is 1 g/cm³.
- ✓ However, the same egg floats in salty or sugar solution because salt or sugar solution has greater density than the egg.
- ✓ On the other hand, **a rotten egg floats on water since its average density is less than the density of water due to accumulation of air inside the rotten egg.**
- ✓ Average density explains why a ship made of steel floats on water.
- ✓ **A ship is composed of air in between steel parts which makes the average density less than the density of water.**

UNIT 3: SPECIFIC HEAT CAPACITY

Heat capacity

Heat capacity of a substance is the heat energy required to raise the temperature of a substance by 1 K.

Mathematically,

$$\text{Heat capacity (C)} = \frac{\text{Amount of Heat supplied (Q)}}{\text{Temperature change } (\Delta\theta)}$$

The SI unit of heat capacity is joule per kelvin (J/K)

Example 4.2

Calculate the quantity of heat required to raise the temperature of a metal block of capacity of 520 J/K from 9°C to 39°C.

Solution

Quantity of heat Q = Heat capacity $\times$ temperature change

$$\begin{aligned} Q &= C \times \Delta\theta \\ &= 520 \text{ J} \times (39 - 9) \text{ K} \\ &= 15\,600 \text{ J} \end{aligned}$$

Example 4.3

The quantity of heat required to raise the temperature of water from 10°C to 65°C is 6 200 J. Calculate the heat capacity of water.

Solution

$$\begin{aligned} Q = C \Delta\theta &\longrightarrow C = \frac{Q}{\Delta\theta} \\ &= \frac{6\,200 \text{ J}}{(65-10) \text{ K}} \\ &= 112.73 \text{ J/K} \end{aligned}$$

The heat capacity of water is 112.73 J/K

Specific heat capacity

In the equation, $Q = mc\Delta\theta$, specific heat capacity $Q = c$, if $m = 1 \text{ kg}$ and $\Delta\theta = 1 \text{ K}$.

When the mass of the substance is 1 kg (i.e. $m = 1 \text{ kg}$) and the change in temperature is 1 K (i.e. $\Delta\theta = 1 \text{ K}$), then $Q = c$ and c is referred to as the specific heat capacity of the substance.

The specific heat capacity, c of a substance is defined as the heat energy required to change the temperature of a substance of mass 1 kg by 1 kelvin .

$$c = \frac{Q}{m\Delta\theta}$$

Therefore,

Quantity of heat = mass $\times$ specific heat capacity $\times$ temperature change

$$Q = mc\Delta\theta$$

Where, $\Delta\theta$ = final temperature – initial temperature

The SI unit of specific heat capacity is **joule per kilogram per kelvin (J/kgK)**.

Example 4.4

Calculate the heat energy required to raise the temperature of 2.5 kg of aluminium from 20°C to 40°C , if the specific heat capacity of aluminium is 900 J/kgK .

Solution

Heat energy required = mass $\times$ specific heat capacity $\times$ temperature change

$$\begin{aligned} Q &= mc\Delta\theta \\ &= 2.5 \times 900 \times (40 - 20) \\ &= 45\,000 \text{ J} \end{aligned}$$

Comparison of specific heat capacities of the three states of matter

- ✓ Substances have different heat capacities.
- ✓ Solids require a lot of energy to melt than liquids and gases.
- ✓ This means that solids have higher specific heat capacity than liquids and gases.
- ✓ Gases have the lowest specific heat capacity.

4.3 Application of specific heat capacity

Specific heat capacity has many application in our daily life. The following are a few examples:

- a) A material with high specific heat capacity absorbs a lot of heat with only a small rise in temperature. This accounts for the efficiency of water as a coolant in a car radiator and of hydrogen gas in enclosed electric generators.
- b) Substances with low specific heat capacities are quickly heated up; they experience a big change in temperature after a small amount of heat energy. For this reason, they are used to make cooking utensils such as frying pans, pots and kettles.
- c) Sensitive thermometers are made from materials with low specific heat capacity in order to detect and accurately show rapid change in temperature, even for small amounts of heat energy.
- d) Materials with high specific heat capacity are suitable as for making handles of heating devices like kettles, pans and oven covers. This is because they do not get very hot and easily when they absorb high amounts of heat energy.
- e) Water consumed by human beings regulates the temperature of the body since water has a high specific heat capacity.
- f) Sea water is cold during day time when the temperature are high because water has a high specific heat capacity. At night, the temperature of water is high and hot air of low density rises from the sea allowing cold air of high density blow from land to the sea called land breeze.

Worked examples:

1) 5000 J of heat is added to water in a pot to raise its temperature by 20°C . Calculate the heat capacity of the water in the pot (**250 J/ $^{\circ}\text{C}$**)

2)The heat capacity of a certain mass of glass is 670J/ $^{\circ}\text{C}$.

a) What will be the raise in temperature if 2010J of heat energy was supplied? (**3 $^{\circ}\text{C}$**)

b) How much heat energy would be required to raise its temperature by 5°C ? (**3350J**)

UNIT 4: HEAT TRANSFER

- ✓ **Heat** is a form of energy which passes from a body of high temperature to a body of low temperature.
- ✓ The flow of heat is mainly from a hotter substance to a cooler one because of their difference in temperature.
- ✓ The SI Unit of heat energy is **joule (J)**
- ✓ There are three methods of heat transfer – **conduction, convection and radiation.**

Conduction

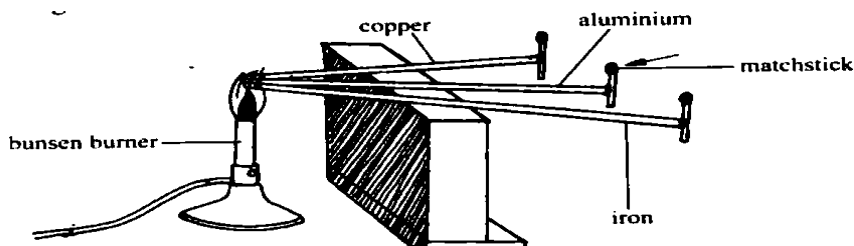
- ✓ Conduction is the transfer of heat from one substance to another that are in direct contact.
- ✓ It can also refer to the movement of heat from molecule to molecule through a material.
- ✓ In conduction, energy transfer takes place by vibration of the molecules.
- ✓ There is no actual visible movement of the heated particles.
- ✓ when you touch a cold object, you feel cold. Heat is transferred from your hands to the object.
- ✓ **Conduction** occurs mainly in solids.

Good conductors and poor conductors

- ✓ A substance which has the ability to transfer heat through itself easily is called **a conductor**. Most metals conduct heat well and are called **good conductors of heat**.
- ✓ Wood, glass, cork and fabrics do not conduct heat.
- ✓ They are poor conductors of heat.
- ✓ Most liquids and gases are poor conductors of heat (**insulators**).

Comparing rate of conduction of heat by different substances

- ✓ Take three rods (copper, aluminium and iron) of the same length and thickness.
- ✓ Fix a matchstick to one end of each rod using a little melted wax
- ✓ Place the rods on a tripod stand and heat the free ends with a burner
- ✓ Observe what happens
- ✓ The matchstick falls off from the copper rod first then aluminium and finally from the iron rod.
- ✓ Copper is the best conductor of heat followed by aluminium and then iron. The matchsticks do not fall off at the same time. Different materials conduct heat at different rates.



Convection

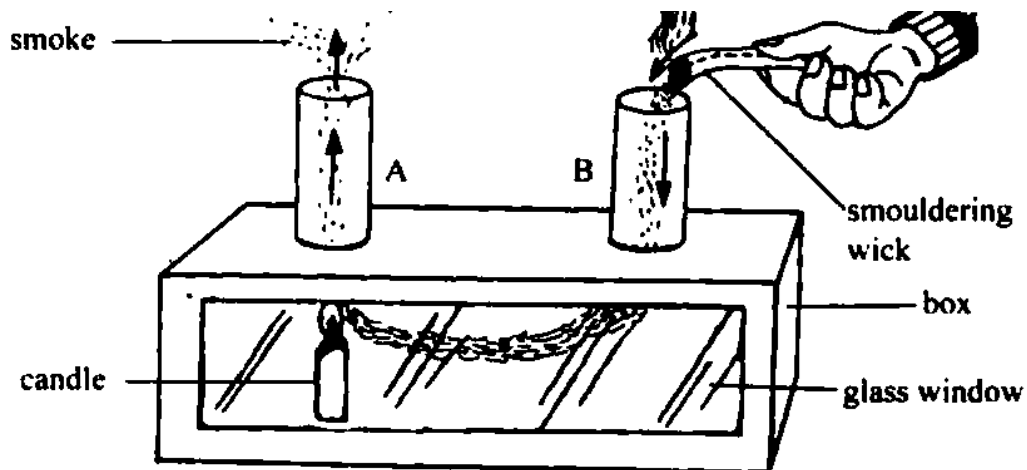
- ✓ Convection is the transfer of heat in a fluid from high to lower temperature region by convection current.
- ✓ **Convection current** is the flow of fluid caused by a change in density, in which the whole medium moves and carries heat energy with it.
- ✓ Convection is an important heat transfer in liquids and gases (fluids) because the particles of gases and liquids move.
- ✓ Convection does not occur in solids because their particles do not move.

i) Convection in Liquids:

- ✓ When you heat a container filled with a liquid (water) added with a crystal of potassium permanganate, the upward movement of water together with potassium permanganate is observed as in figure above.
- ✓ When water is heated, the water at the bottom of the beaker becomes lighter because it is less dense.
- ✓ The warm water rises to the surface and cooler water which is denser than heated water sinks to the bottom of the beaker and is heated up in turn.
- ✓ Convection current is set up in which hot liquids rise and cold liquids sink to the bottom.
- ✓ It is this circulation of the liquid (convection current) that transfers heat from the heat source to other parts.

ii) Convection in Gases:

- ✓ Convection currents also occur in air and gases.
- ✓ During a hot day, the sun heats the road and other surfaces.
- ✓ A column of warm air above the road, expands, become less dense and then rises to float as convection current.
- ✓ **An experiment with a convection box helps to show that heat can also transfer by convection through gases like air as seen in figure below.**



- ✓ Light a candle and place it at **the base of the glass chimney** as shown in figure above. Hold a smoke straw over the second chimney.
- ✓ The air above the flame gets heated up, expands, becomes less dense and then rises up the chimney directly over the candle flame.
- ✓ The cool, denser air sinks through the second chimney to replace the risen hot air.
- ✓ The cool air falls through the second chimney together with smoke from the smoke straw making air currents visible.
- ✓ The smoke reveals the direction of convection currents.
- ✓ **Hot water systems, an engine cooling system and land and sea breezes are some examples of convection currents.**

Radiation

- ✓ **Radiation** is the transfer of heat through a vacuum.
- ✓ **These observations explain heat transfer by radiation:** When you stand in the sun, you will feel warm because of heat is radiated from the sun to your body by radiation.
- ✓ Heat has not transferred by conduction because air is a poor conductor of heat and there is a vacuum between the sun and the earth.

Applications of heat transfer in everyday life

Conduction of heat is used in the following situations:

- a) **Cooking pots:** Cooking pots are made of metals to conduct heat faster. Their handles are made of plastic or wood because it is an insulator
- b) **Tea cups:** Most tea cups are made of insulators

- c) **Clothing:** Air trapped next to the skin reduces heat loss or gain on a very cold day. Likewise, it minimizes heat gain on a hot day. Note that air is a bad conductor of heat.
- d) **Wall insulation:** In cold weather, countries minimize heat loss or gain by making the wall of the house with two sets of bricks with a gap between them. The gap is filled with an insulating material.

ii) Convection currents are observed in land and sea breezes

a) Land breeze –

- ✓ At night, both the land and the sea lose heat. The land loses heat much faster than the sea, having lower specific heat capacity than the sea.
- ✓ This causes the warm air above the sea to rise because it is less dense. Cold air blows from the land to the sea to replace the risen air.

b) Sea breeze

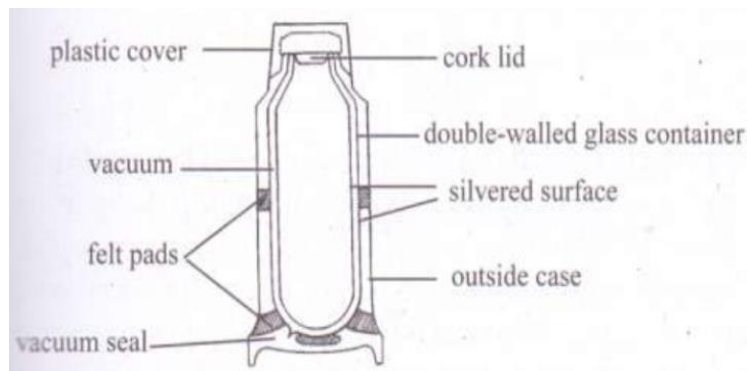
- ✓ This occurs during the day when the land heats up faster than the sea.
- ✓ This causes air above the land to rise because it becomes less dense, allowing the cold dense air over the sea to blow to the land.
- ✓ This creates a sea breeze in the daytime because cold dense air blows from the sea to the land.

iii) Radiation of heat is used in the following situations:

a) A thermos or vacuum flask

- ✓ keeps hot liquids hot and cold liquids cold.
- ✓ It is designed to reduce heat loss by conduction, convection and radiation.
- ✓ The wall of the flask is made of double layer of silvered glass.
- ✓ The space between the double layers is **shiny** / **silvered** to prevent heat entering the flask by radiation (or to reduce heat loss by radiation).

- ✓ The cork lid is poor conductor of heat. Due to design problems, the flask slowly loses heat as seen in figure below



- b)** Houses in the hotter areas on earth are painted white to reflect heat and keep the inside of the house cool.
- c)** Wearing white clothes on a hot day leaves the body cooler than wearing black clothes
- d)** The inside of a food warmer is made shiny to prevent a lot of heat from escaping and the food stays warm..

UNIT 5: POWER AND MACHINES

Power

- ✓ **Power** is defined as the rate of doing work.
- ✓ It can also be defined as the rate at which energy is changed.
- ✓ Some people have more energy and do work faster than others. They finish their work in a shorter time.
- ✓ We say that these people are more powerful.
- ✓ The symbol for power is **P**. **Mathematically**,
- ✓ **Power (P) = work done (J) / time taken (s)**

Basing on energy changed, Power = energy changed / time taken

- ✓ **Work or energy** is measured in **joules** and **time** is measured in **seconds**.
- ✓ **Power is measured in J/s.**
- ✓ **However**, the **SI Unit** of power is **watt (W)**. **1W = 1 J/s.**
- ✓ **Work done = Force x distance moved in the direction of force.** **Rate** measures how fast something can be done.

Calculations on power

- ✓ People and machines have a certain value of power when they do work.
- ✓ Different persons have different rates at which their muscles use energy.

Worked examples:

1)A treadle pump raises water from a well to height of 30 metres. It pumps 100 dm³ in 1 minute.

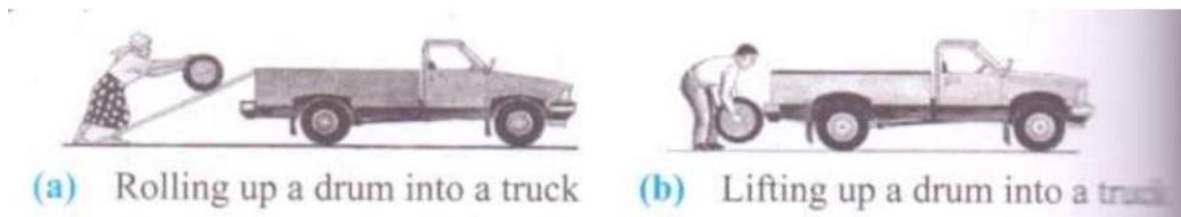
Calculate the power of the pump (**500W**)

2) A student of mass 45 Kg runs up a flight of 40 steps in a stair case each 15 cm in 12 seconds.

Calculate the power of the student (**225 W**)

Machines

- ✓ Machines are devices that make work easier.
- ✓ For example in loading an oil drum onto a truck, it is easier to roll it up an inclined plane than lifting it up onto the truck.



a. Rolling up a drum into a truck

b. Lifting up a drum into a truck

- ✓ A machine may be defined as any device that facilitates a force applied at one point to overcome another force at different point in the system.
- ✓ In mechanical machines the force that is applied is called the effort (E) and the force the machine must overcome is called the load (L).
- ✓ A machine is therefore a device which makes work easier.

Terms used in machines

- ❖ Machines can be described using several terms.
- ❖ These terms are mechanical advantage, velocity ratio, efficiency and the characteristics of a machine.

Mechanical Advantage of a Machine (MA)

- ✓ The number of times a machine magnifies the effort (force applied) is **MA**.
- ✓ In other words, the **MA** of any machine refers to the number of times the load is greater than the effort.
- ✓ The ratio of Load to Effort is called **mechanical advantage (MA)**.
- ✓ Mathematically, **MA = Load / effort**.
- ✓ It has no units since it is a ratio of two forces.
- ✓ Where MA is greater than 1, it means the machine is designed to overcome a load which is greater than the effort. An example is a car jack used to lift a motor car.
- ✓ Where MA is less than 1, it means the machine is designed so that the effort used is greater the load. For example, a bicycle has a MA of less than 1.

Example:

A man applied a force of 300N to a body of mass 300 Kg on an inclined plane. Calculate the mechanical advantage of machine.

Solution

$$MA = L / E = 3000 \text{ N} / 300 \text{ N} = 10$$

Velocity Ratio (VR) of Machines

- ✓ Velocity ratio (VR) of a machine is the ratio of the velocity of the effort to the velocity of the load.

VR = Velocity of the effort / Velocity of the load

= **Effort distance / load distance**. Hence velocity ratio may be defined as the number of times the effort moves further than the load. Just like MA, VR has no units.

Efficiency of the Machine (E)

- ✓ **The efficiency of a machine** is defined as the ratio of the work output to the work input expressed as a percentage.
- ✓ **Efficiency** is also the ratio of mechanical advantage to velocity ratio.

Efficiency Formula

Efficiency = work (energy) output / work (energy) input x 100 %

- ✓ Where work done by the effort = **work input**, and work done on the load = **work out put**.
- ✓ Work input is always greater than work output since work has to be done against friction, air resistance and weight.

Or Efficiency = MA / VR x 100 % (iv)

Examples

1. A machine; the load moves 2 m when the effort moves 8 m. If an effort of 20 N is used to raise a load of 60 N, what is the efficiency of the machine?

Solution

$$\text{Efficiency} = (M.A / V.R) \times 100 \quad M.A = \text{load/effort} = 60/20 = 3$$

$$V.R = DE / DL = 8/2 = 4$$

$$\text{Efficiency} = \frac{3}{4} \times 100 = 75\%$$

Simple Machines

- a) **Levers** – this is a simple machine whose operation relies on the principle of moments
- b) **Pulleys** – this is a wheel with a grooved rim used for lifting heavy loads to high levels.

The can be used as a single fixed pulley, or as a block-and-tackle system.

$$M.A = \text{Load/ Effort}$$

$$V.R = \text{no. of pulleys/ no. of strings supporting the load}$$

Example

A block and tackle system has 3 pulleys in the upper fixed block and two in the lower moveable block. What load can be lifted by an effort of 200 N if the efficiency of the machine is 60%?

Solution

$$V.R = \text{total number of pulleys} = 5$$

$$\text{Efficiency} = (M.A / V.R) \times 100 = 60\%$$

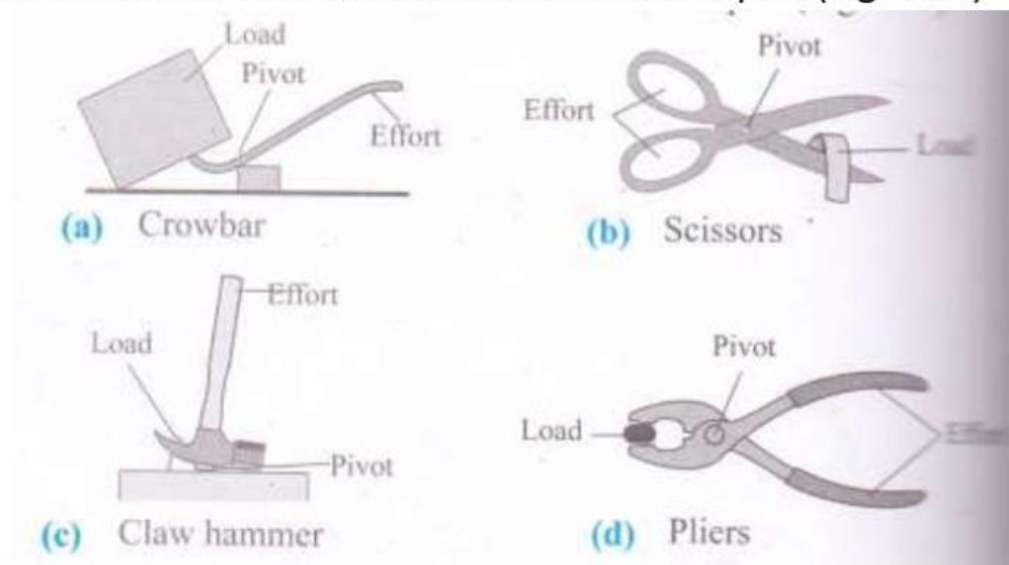
$$0.6 = M.A/ 5 \Rightarrow 3, \text{ but } M.A = \text{Load/Effort}$$

$$\text{Therefore, load} = 3 \times 200 = 600 \text{ N}$$

Types of Lever systems

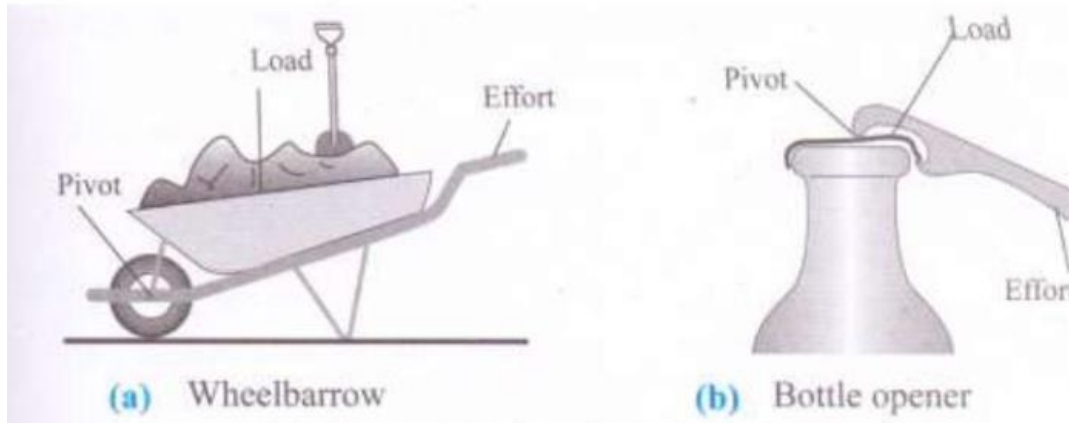
There are three types of levers. The difference between these types depend on the position of the pivot with respect to the load and the effort.

1. The pivot is in between the load and the effort. Examples (Fig. 6.23).



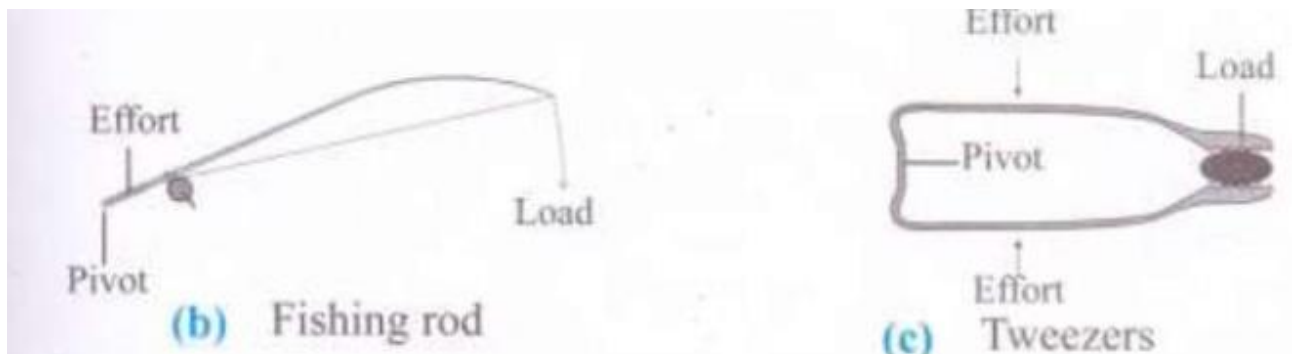
- ✓ In this class, the lever changes the direction of force, meaning that load acts in one direction with the effort being applied in the opposite direction.

ii) Levers of the second order - Levers of the second order have the load between the fulcrum and the effort as seen in figure. When the load is nearer to the fulcrum, the effort needed to lift the load will be less. Examples of second class levers are a wheelbarrow, a nut cracker and a bottle opener.



iii) Third order lever

- ✓ Levers of the third order have the effort applied between the load and fulcrum, as seen in figure. Examples of third order levers are long broom, tongs / tweezers, human fore arm, fishing rod and crow hammer.

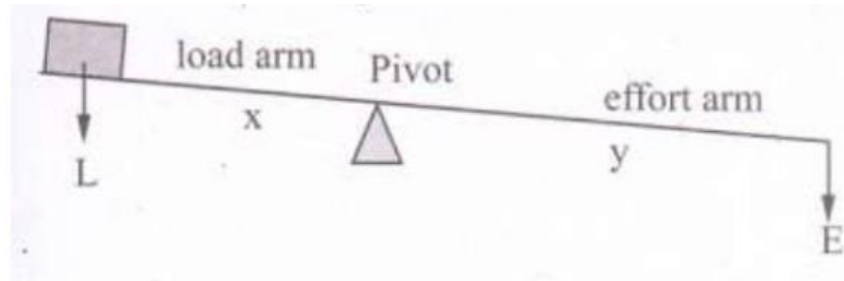


Note that

- ✓ the first and second order levers are referred to as 'force multipliers' since less effort is used in lifting or overcoming a heavy load.
- ✓ A third order is a distance multiplier since the load covers a long distance with more effort exerted.

Principle of Moments in levers

- ✓ Levers use the principle of moments.
- ✓ **The principle of moments / the law of orders** states that the sum of clockwise moments about a point is equal to the sum of anti-clockwise moments at the same point at equilibrium.



- ✓ **Where d_1 = load distance (load arm) and d_2 = effort distance (effort arm)**
- ✓ At balance: Sum of clockwise moments = Sum of anti-clockwise moments. Load x load arm = Effort x effort arm. **Mathematically, $d_2 \times E = d_1 \times L$**
- ✓ Load arm is the distance between the load and fulcrum and effort arm is the distance between effort and the pivot.
- ✓ **$MA = \text{load} / \text{effort} = \text{effort arm} / \text{load arm}$**

Worked example:

Calculate the amount of load and MA in the lever system in figure below (**Load = 250N and MA = 5**).

Inclined Plane

An inclined plane

- ✓ Is a machine which is in form of a slope or hill. It is a flat surface placed at an angle with one end higher than the other.
- ✓ It is a machine because it allows one to use less effort than the load.
- ✓ The inclined plane is used as an aid for raising or lowering the load. Simply put, it is easier to move a heavy object up an inclined plane than to move it vertically upwards.
- ✓ Some examples of incline planes are a winding road up a hill, a wedge, a knife, sharp pointed fronts of ships, airplanes, cars and fish, screw nail.

Ways of increasing the MA of the inclined plane

- ❖ Using rollers instead of sliding reduces the friction between the load and the inclined plane.
- ❖ Using the longer planks reduces the height above which the load has to be lifted.
 - ❖ Using a smooth plank requires less effort in pulling an object along the plane.
 - ❖ All these decrease effort applied in moving the object along the inclined plane hence MA increases.

UNIT 6: ELECTROSTATICS

- ✓ **Electrostatics** is the study of static charges; that is, charges that are at rest.
- ✓ The charges are stationary or do not move.
- ✓ Electrostatics is about charges in insulators.
- ✓ Static electricity also refers to the flow of electrons in insulators.
- ✓ **The negative and positive charges** are two types of static charges.
- ✓ **A charge** is created when electrons move from one object to another. Atoms are made up of three sub-atomic particles:

- **protons (+ve) charge**

- **electrons (-ve) charge**

- **neutrons (have no charge)**

- ✓ **An electric charge** is the physical property of matter that causes it to experience an electrostatic force when close to other electrically charged matter.
- ✓ The symbol for electric charge is **Q**.
- ✓ The SI unit of an electric charge is **Coulomb (C)**.

The Law of Charges

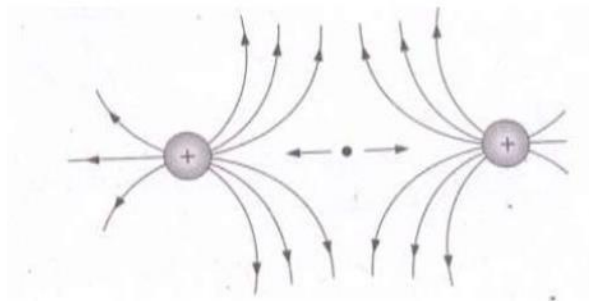
- ✓ The law of charges states that like charges repel and unlike charges attract.
- ✓ For example, a negatively charged polythene rod and a positively charged cellulose acetate **attract each other** when freely suspended (held) from cotton threads and placed near to each other.

Electric field

- ✓ An electric field is the region or space surrounding a charge (or electric lines of forces). Simply put, an electric field is the region where a charged body experiences a force.
- ✓ In this region, another charged body may move away from or towards the charged body producing the electric field.

Electric field patterns

- ✓ In an electric field, the electric force exists at all points.
- ✓ The electric field has direction and is represented using arrow lines



Electrostatic force

The charged plastic pen exerts an attraction of electrostatic force on the pieces of tissue paper, water jet and dry hair.

Factors that affect the magnitude of the electrostatic force between two charged objects

i) Quantity of charge

- ✓ the greater the quantities of charge on the two charged objects the greater the force between them since there will be more charges which may result to repulsion or attraction.

ii) Distance of separation between interacting charges

- ✓ the greater the distance between the two charged objects, the smaller the force and vice versa.
- ✓ For example, when the separation distance is short, the strength of the force of attraction between the two charged objects increases.

Methods of charging Insulators

1. Charging by Friction –

- ✓ This is charging by rubbing two objects together which results in friction. Friction involves rubbing two surfaces against each other.
- ✓ Energy used to produce charges comes from rubbing (friction) a material with a piece of dry woolen cloth.

2. Charging by Contact

- ✓ This refers to charging in which a neutral object (uncharged object) gets the charge of the charged object in contact (touch).
- ✓ When a neutral object, for example, is in contact with a positively charged cellulose acetate, the charged acetate will eventually be neutral and the neutral object will be positively charged.

3. Charging by Induction

- ✓ Charging by induction refers to charging in which a charged object produces an opposite charge to the nearby object.

The Electroscope

- ✓ It is a device which is used to detect charge on an object.
- ✓ Electroscopes are used to detect the presence of electric charge on an object and also to work out the sign of the charge on an object.

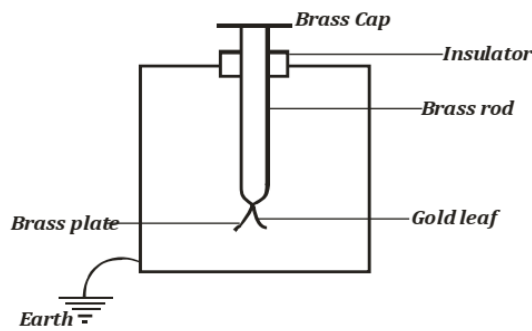
i) Pith ball Electroscope

- ✓ A pith ball is a very light ball that is suspended freely by a thin insulating thread of nylon.
- ✓ The pith ball is touched and charged by contact using a negatively charged polythene rod hence it becomes negatively charged.

- ✓ A pith ball is charged either positive or negative. If charged object is brought closer to the pith ball then the charge can be detected by using the principle of attraction and repulsion.

ii) Gold leaf electroscope

- ✓ The gold leaf has a thin foil of gold leaf attached to a metal plate **Testing type of charge on material using leaf electroscope**, first give the device known type of charge by contact



- ❖ The gold-leaf electroscope is an instrument used for testing positive and negative charges.
- ❖ It consists of a metal (brass) rod to which a thin gold-leaf (or aluminium leaf) is attached.
- ❖ The rod is surmounted by a brass disc or cap and insulated from the metal case.
- ❖ The leaf is protected from outside influences (like drought) by enclosing it in an earthed metal case with glass windows.
- ❖ Generally, the gold-leaf can also be used to test the conducting properties of materials.

Applications of electrostatics

a) Lightning conductor

- ✓ Is used to protect tall building from being destroyed by lightning discharge.
- ✓ A lightning conductor is made up of a thick copper strip.

b) Capacitor – a device used to store charge and energy.

c) Paint Spraying

- ✓ The part to be painted is given a negative charge.
- ✓ The nozzle of paint spray gun is given a positive.
- ✓ Paint droplets become positively charged by friction as they emerge from the nozzle.
- ✓ Therefore, positively charged paint is attracted to the negatively charged part to be paint and hence stick to it, making a very smooth and firm coating.

d) Photocopiers

These use static charge to copy documents.

e) Electrostatic Precipitation (removing dust from smoke)

- ✓ Chimneys and exhaust tubes are filled with two sets of plates – first plate is positively charged and the second is negatively charged.
- ✓ As the mixture of the smoke and dust passes through the chimney, dust particles become positively charged by the first set of plates.

UNIT 7: LIGHT

Light is a form of energy. It enables us to see the surrounding objects. Light itself is not visible but its effect is felt by the eye. For example, the track of light entering a room cannot be seen; but the track becomes visible, if some dust particles are present in the room. In a cinema theatre, light from the projector to the screen is visible due to the dust or smoke moving through the path of light.

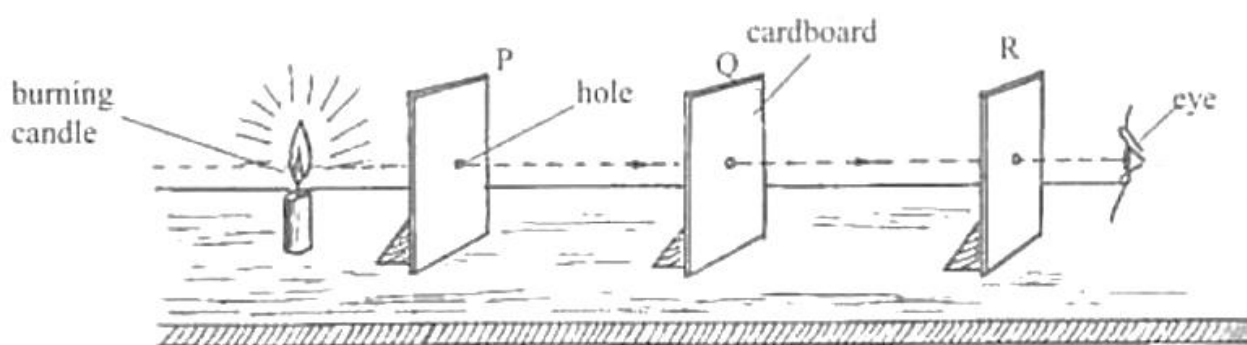
To show that light travels in a straight line.

Apparatus

- Soft board
- A plane mirror
- Plasticine
- White sheet of paper

Procedure

- Take three cardboard P, Q and R of equal sizes mounted on wooden stands.
- Make a small hole at the same height and also at equal distances from the edges on each cardboard. Place the cardboards on a flat surface (Bench). When a thread is passed through these holes, it should be straight and taut. This is to ensure that the three pinholes are along the straight line.
- Remove the string without disturbing the setup of the three cardboards.
- Place a lighted candle in front of the hole in the cardboard P and view from the side of R as shown in Fig. What conclusion can you make about the alignment of the three holes and the manner in which the light travels?



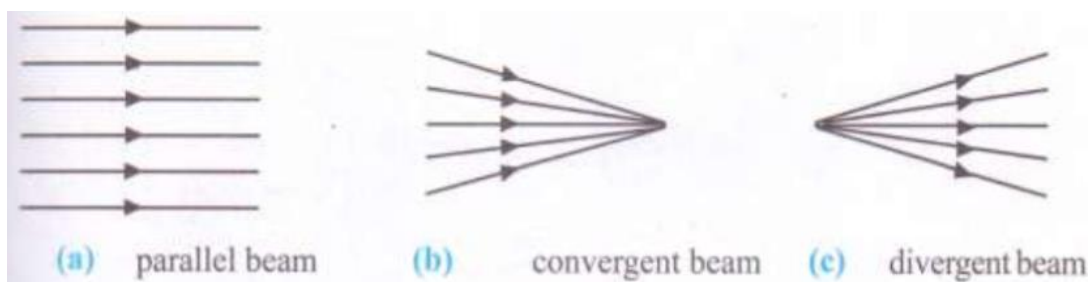
Some useful terms in the study of light

Luminous bodies: These are bodies which emit light on their own. The sun, a candle flame, etc., are known as self-luminous bodies.

Non-luminous bodies: These are bodies which have no light of their own, but become visible in the presence of some luminous bodies e.g. moon, door, tree, etc.

Optical medium: This refers to a substance through which light can pass. A medium can be transparent, opaque or translucent. A medium like water, air, glass through which light can pass freely is said to be transparent.

A ray of light: The path along which light travels in a medium is called a ray. A collection or groups of rays is called a beam. A beam can be parallel, convergent or divergent



Formation of shadows and eclipses

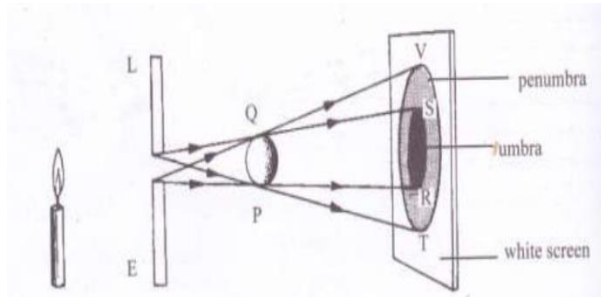
A shadow is a region where light cannot reach.

- ✓ The shadow is formed when light travels in a straight line and some rays of light fail to pass through an opaque object. It is formed on the other side of the opaque object where light cannot reach.
- ✓ The sharpness of the shadow depends on the size of the source. The sources of light can also be grouped into point sources and extended sources.
- ✓ **A point source** is where the light appears to come from a small single point, for example, light from a very small hole.
- ✓ In **extended sources light** comes from so many points of the light source, for example, sun and light from a large bulb.

- ✓ The region of complete darkness is called **umbra**.
- ✓ The shadow formed by a point source has sharp-edges and contains umbra only.

Formation of shadows with an extended source of light

- ✓ A large opening through cardboard forms **an extended source of light** when illuminated with light.
- ✓ An opaque QP placed between LE and a white screen blocks light rays as shown in figure



Eclipses

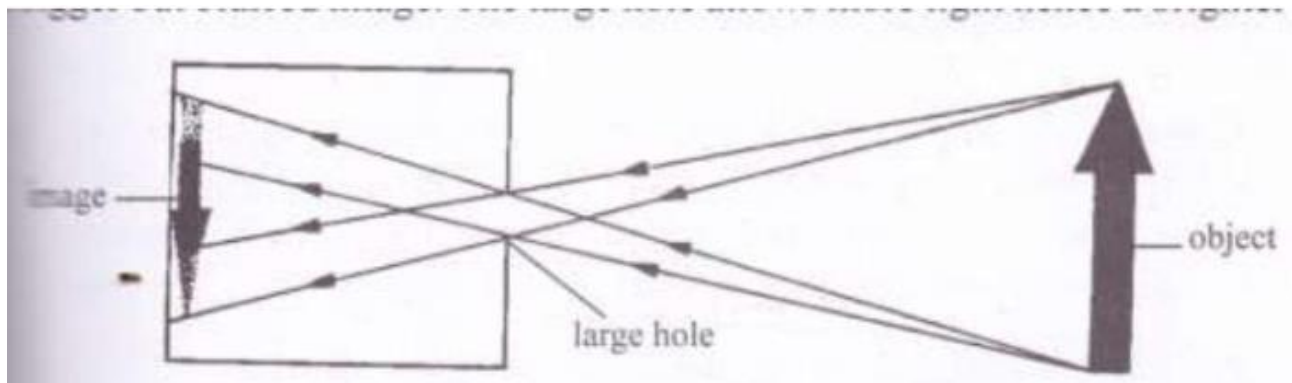
An important example of the formation of shadows is the occurrence of eclipses. The term eclipse means that light is blocked or cut off from region of observation. Two eclipses will be discussed here namely solar eclipse and lunar eclipse.

Solar eclipse

When the moon, revolving around the earth, comes in between the sun and the earth, the shadow of the moon is formed on the earth. This is called the solar eclipse or the eclipse of the sun. Depending on the position of the moon, some parts of the earth lie in the region of umbra and some in the region of penumbra.

Pinhole Camera

- ✓ it consists of a box with a pinhole on one side and translucent screen on the opposite side.
- ✓ It is an instrument that can be used to observe light travelling in a straight line.
- ✓ Light rays from an object, pass through the pinhole and form an image on the screen, as shown in figure below.



- ✓ The pinhole camera forms a **real and inverted image** (a real image is an image that can be formed on a screen while a virtual image is an image that cannot be formed on the screen).
- ✓ **Note that a pinhole camera forms an inverted image (upside down) because light rays from the top and bottom of the object cross as they pass through the pinhole.** A pinhole camera has a large depth of field or a large depth of focus.

Magnification produced by a pinhole camera

- ✓ The term linear magnification refers to how big or how small the image is as compared to the object.
- ✓ Magnification is defined as the ratio of the height of the image to the height of the object.

Magnification (m) = height of the image (IM) / height of the object (OB)

$$m = IM / OB \dots\dots\dots (1)$$

Worked example:

- ✓ A pinhole camera of length 20cm is used to view the image of a tree of height 12m which is 40 m away from the pinhole. Calculate the height of the image of the tree obtained on the screen.

Properties of light

- i) reflection of light
- ii) refraction of light
- iii) dispersion
- iv) diffraction
- v) interference of light

Look at yourself in a plane mirror. If your shirt or blouse pocket is on the left side, your image appears to have a pocket on the right hand side. However, the image is upright and of the same size.

From the above observations, we can summarize the characteristics of images formed by plane mirrors as follows:

1. The size of the image is equal to the size of the object.
2. The image is erect.
3. The image is virtual
4. The distance of the image behind the mirror is equal to the distance of the object in front of the mirror.
5. The image is laterally inverted.

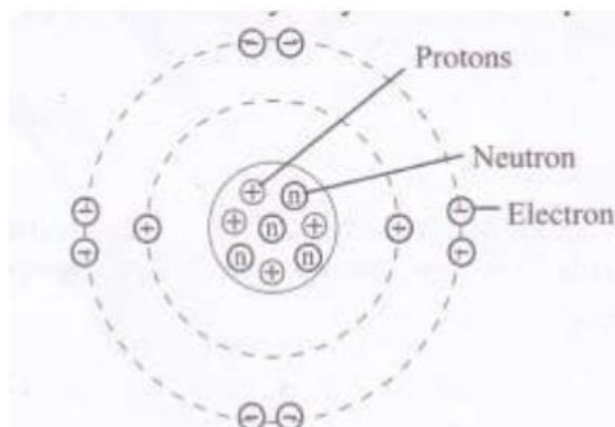
Applications of Reflection of Light

- i) Reflection enables us to see the non- luminous objects. The surface of non- luminous objects reflects the rays of light into our eyes.
- ii) Reflection enables us see our images on the mirror.
- iii) Reflection is also used to see our images on the mirror
- iv) Reflection are also used in reflecting telescope. In a telescope, the rays of light are reflected back by the concave mirror in order to illuminate the object.
- v) Reflection is also used in kaleidoscope (mirrorscope) which is used to see different patterns of light.
- vi) **Periscope** – it is a device used to see objects over heads of people and around corners of obstacles. Submarines use periscopes to see or monitor what is happening above the water.

UNIT 8: INTRODUCTION TO NUCLEAR PHYSICS

Structure of an atom

In Junior 1, you learnt that matter is made up of tiny particles called atoms. For a long time, scientific activities thought that atoms are the smallest building blocks of matter and they could not be subdivided further. In 1897, a particle smaller than an atom called electron was discovered. Today they have a better picture of an atom.



An atom is made up of two parts. The central core called the nucleus where protons + and neutrons n are closely packed and an outer orbit where electrons go round its nucleus. They carry a **negative** charge. Protons carry a **positive** charge. Neutrons carry no charge. The number of protons and neutrons are issued in as an atom and hence an atom is always neutral or uncharged.

Charge number or atomic number (Z) is the number of protons inside its nucleus
Mass number or atomic weight (A) is the number of protons and neutrons inside the nucleus. Protons and neutrons are called the nucleus.

A

X

Z

Where X – as symbol for the element

A – atomic mass

Z – atomic number

Example 9.1

Given that Lithium atom ${}^7_3\text{Li}$. Identify the charge number, mass number and the number of neutrons.

Solution

The charge number is 3, i.e. the number of protons inside the nucleus

The mass number is 7, i.e. the number of protons and neutrons inside the nucleus.

The number of neutrons is equal to $7 - 3 = 4$

Isotopes

Isotopes of an element are atoms which have the same number of protons but different number of neutrons i.e. isotopes are atoms with the same atomic number but different mass number.

Example 9.3

Explain why ${}^1_1\text{H}$, ${}^2_1\text{H}$, ${}^3_1\text{H}$ are isotopes of hydrogen atom.

Solution

${}^1_1\text{H}$, ${}^2_1\text{H}$ and ${}^3_1\text{H}$ have the same number of protons ($z = 1$) but different mass number i.e. different number of neutrons.

In ${}^1_1\text{H}$ there are no neutrons. In ${}^2_1\text{H}$ hence 1 neutron and in ${}^3_1\text{H}$ there are 2 neutrons

Radioactivity

- ✓ It is the process by which an unstable nucleus spontaneously disintegrates (decays / breaks up), emitting sub-atomic particles or radiations.
- ✓ **Radioactivity** is the breaking apart of the nucleus of an atom. Some nuclei are stable while others are unstable. The nuclei that are unstable are the ones that disintegrate.
- ✓ **Radiation** is the emission of energy or particles from a material.
- ✓ **Nuclear radiation** is the emission of sub-atomic particles and rays of energy from the nucleus of an atom.

Types of nuclear radiation

Radioactivity involves three types of radiations or emissions namely: **Alpha (α) particles**, **Beta (β) particles** and **Gamma (γ) rays**.

1. Alpha emission

- ✓ Alpha emission is a loss of alpha radiation by a radioactive element.
- ✓ Alpha particle is the nucleus of a helium atom with atomic number 2 and mass number 4.
- ✓ Its nuclear symbol is: ${}^4\text{He}$ or ${}^4\alpha$.
- ✓ Alpha particle does not contain electrons unlike helium atom. It has a charge of +2 having 2 protons and mass number of 4 due to presence of 2 neutrons and 2 protons.

Properties of Alpha particle

- i) It is a nucleus of helium atom
- ii) It has a net positive charge
- iii) It attracts electrons from nearby atom
- iv) It has little penetrating power because it is heavy
- v) It is readily absorbed by air, paper and even surface of the body

2. Beta (β) Emission

- ✓ During beta emission, a radioactive substance decays by releasing beta radiation.
- ✓ Beta particle is the fast-moving electron emitted from the nucleus of a radioactive element.
- ✓ It has low mass and is negatively charged. Lack of mass makes beta particles to be light. The source of a beta particle is Strontium (Sr) and its nuclear notation is ${}^0\text{e}$ or ${}^0\beta$.

Properties of Beta Particles

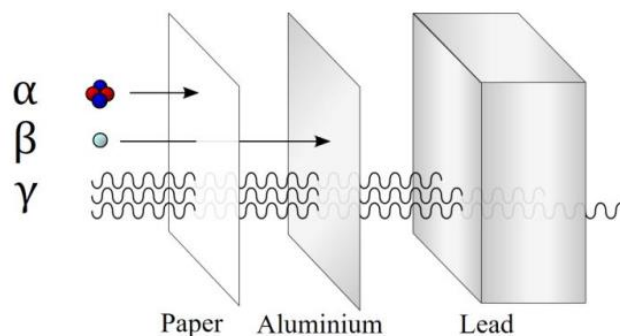
- ❖ These are electrons, carry a negative charge
- ❖ They have greater penetration power than alpha particles because they are very tiny
- ❖ They can be stopped (absorbed) by few mm of Aluminum or tin foil
- ❖ They cause less ionization of air
- ❖ They are deflected much in both magnetic and electric fields because they are tiny.

3. Gamma (γ) emission

- ✓ Gamma emission is simply the release of gamma rays.
- ✓ Gamma rays are electromagnetic waves that are shorter in wavelength than X – rays.
- ✓ Gamma rays are also high energy emitted from the nucleus of a radioactive substance.

Properties of Gamma rays

- ❖ They are transverse waves in nature
- ❖ They have highest penetrating power when compared to alpha and beta particles
- ❖ They cause little ionization of air particles because they have no mass and have highest speed
- ❖ They show no deflection in either magnetic or electric fields since they are not charged
- ❖ They can be stopped or absorbed by very thick lead or concrete



Examples of materials which will stop alpha beta and gamma radiation

Uses of radioactivity / nuclear radiations

1. In Agriculture

- a) Pest control-** sterilizing male insects to control insect population. When males are exposed to gamma radiations, mating with the females produce infertile eggs and no new pests are born.

- c) **Used to study the uptake of fertilizers by plant roots to the leaves** by adding a radioactive isotope (tracer: P-32) to the soil water and trace its flow by a detector.

2. Medical Uses

- (a) **As a radiotherapy** – treatment of cancer using gamma rays. Gamma rays emitted from a strong source of Cobalt-60 can be used to kill cancer.

- (b) **Sterilizing medical Equipment** – Medical equipment (syringes) are sterilized by irradiating them with gamma rays because they (rays) kill all germs.

2. Carbon dating.

This is the process of estimating the age of dead plants and animals by measuring the activity of C-14 present in them.

3. Power generation-

- ✓ In nuclear power stations, nuclear fission is used in a nuclear plant to release heat energy that is used to boil water.
- ✓ The steam then turns turbines to produce electricity.